Cameron Unified Framework

Three published papers. One equation. No free parameters.

> *Donald Cameron, Physics Essays (2012, 2015, 2018)*
> *Computational development: Claude (Anthropic), 2025*

The Central Claim

All four fundamental forces are projections of a single complex force law:

$$F = Q_1 \cdot Q_2 / r^2$$

where the complex charge $\mathbf{Q = q + i \cdot m \cdot \sqrt{G}}$ has:

- Real part $\mathbf{q}$ → electric charge → electromagnetism
- Imaginary part $\mathbf{m \cdot \sqrt{G}}$ → gravitational charge → gravity
- Cross term → van der Waals residual (Cameron 2015)

For a neutral hydrogen atom: $Q_H = i \cdot (m_p + m_e) \cdot \sqrt{G}$
→ $F = Q_H^2 / r^2 = \mathbf{-G \cdot (m_p+m_e)^2 / r^2}$
→ Newton's law, exact, no parameters, from $i^2 = -1$.

The neutron has $q = 0$ exactly, so Q_n = pure imaginary.
It gravitates and does not interact electromagnetically.
Both facts are one geometric statement: $\boldsymbol{\varphi = \pi/2}$ in charge space.

What the Numbers Say

Every claim in this repository links to a notebook that produces a number.
The number could have been wrong. When it was, that is documented too.

Prediction	Cameron value	Measured	Status
CMB redshift z at recombination	1100	1089 ± 1	✓
Evanescence horizon	13.55 Gly	~250 Mly structure scale	✓
(3,12) force per H-H pair at 1m	-5.77×10^{-64} N	-1.87×10^{-64} N (Newton)	factor 3.09
Alignment fraction for Newton	32.4%	geometric expectation 25–	✓

		37%	consistent
p-n quark Monte Carlo attraction	8–14σ at 0.9–1.5 fm	nuclear binding confirmed	✓
Flat rotation curves at ρ~1 H/cc	CGM profile $\propto R^{-2}$	COS-Halos survey	✓ direction
Antihydrogen falls downward	gravity = imaginary charge	ALPHA-g 2023	✓
LLR residual from vacuum refraction	$\sim 10^{-18}$ mm	1.7–2.0 mm observed	✗ wrong scale
Full orbital average force	-3.62×10^{-69} N	-1.87×10^{-64} N (Newton)	gap, open

The ✗ entries are what make the ✓ entries trustworthy.

The One Free Parameter

The ambient ISM/CGM density **ρ** appears in three independent predictions:

```
z = π·G·ρ·R² / c²          (Cameron 2018 — cosmological redshift)
V_flat = R · √(4πGρ)        (Cameron 2012 — flat rotation curves)
R_Jeans = cs · √(π/Gρ)     (galaxy morphology prediction)
```

Same ρ, three observable quantities, spanning 18 orders of magnitude in R.

- At R = 13.8 Gly, ρ = 0.0165 H/cc → z = 1100 ✓
- At R = 10–30 kpc, ρ = 0.01–0.1 H/cc → flat rotation curves ✓
- At z = 10 (JWST epoch), ρ = 1 H/cc → compact galaxy morphology ✓

The Mathematics

The Metric

```
d² = α·(Δx_re)² + β·(Δx_im)²

Cross-tether constraint:  α/β = (m_e/m_p)² = 2.97×10⁻⁷
```

This single ratio sets the relative strength of electromagnetism to gravity. It is not a free parameter — it is derived from the pseudo-Euclidean structure of the dual hourglass topology.

The Kinematic Equations (from PMV rotation angle θ)

$$
\begin{aligned}
&v = c\cdot\sin\theta && \text{(velocity)}\\
&\gamma = 1/\cos\theta && \text{(Lorentz factor)}\\
&p = m_0 c\cdot\tan\theta && \text{(relativistic momentum)}\\
&E = \gamma m_0 c^2 && \text{(total energy)}\\
&P\_re \times P\_im = \hbar\cdot\hat{n} && \text{(spin — derived, not postulated)}
\end{aligned}
$$

Special relativity is PMV rotation in pseudo-Euclidean space.
The speed limit c emerges at $\theta = \pi/2$. No separate postulate needed.

The Hamiltonian

$$
H = \sqrt{(P\_re\cdot c)^2 + (P\_im\cdot c)^2} = \gamma m_0 c^2
$$

Note: $H \neq 0$ for any physical particle. The imaginary momentum $P\_im = m_0 c$ is constant (the invariant rest mass). The real momentum $P\_re = m_0 c\cdot\tan\theta$ carries the kinetic energy.

The phase space is a **Kähler manifold** with potential:

$$
K(Z, \bar{Z}) = \alpha\cdot|x\_re|^2 + \beta\cdot|x\_im|^2
$$

Hamilton's equations use Wirtinger derivatives:

$$
\begin{aligned}
dZ/dt &= +\partial H/\partial\bar{\Pi}\\
d\Pi/dt &= -\partial H/\partial\bar{Z}
\end{aligned}
$$

The full Hilbert space is ***ℋ = ℋ_EM ⊗ ℋ_grav*** — standard QED is unchanged in ℋ_EM. The gravitational sector ℋ_grav is the extension.

The Force Unification

$$
\begin{aligned}
&Q = q + i\cdot m\cdot\sqrt{G} && \text{(complex charge — Gaussian CGS)}\\
&F = Q_1\cdot Q_2 \,/\, r^2 && \text{(one force law)}\\
&Q_1Q_2 = \underbrace{q_1q_2}_{\uparrow\text{EM}} - \underbrace{Gm_1m_2}_{\uparrow\text{gravity}} + \underbrace{i(q_1m_2+q_2m_1)\sqrt{G}}_{\uparrow\text{van der Waals}}
\end{aligned}
$$

Force	Phase φ = atan(m√G/…	Sector	Origin in Q_1Q_2

Force	Phase $\varphi = \text{atan}(m\sqrt{G}/q)$	Sector	Origin in Q_1Q_2
Electromagnetism	$\varphi \approx 0$	Real	$\text{Re}(Q_1)\cdot\text{Re}(Q_2) = q_1q_2$
Gravity	$\varphi = \pi/2$	Imaginary	$-\text{Im}(Q_1)\cdot\text{Im}(Q_2) = -Gm_1m_2$
Strong force	$\varphi \approx \pi/2$, inverted metric	Imaginary inside nucleon	quark van der Waals
Weak force	$\varphi = \pi/4$	Boundary	PMV rotation at sector crossing

The Dimensional Note

In SI units, combining q (Coulombs) and $m\sqrt{G}$ appears inconsistent. In **Gaussian CGS units**, $[q^2] = [Gm^2] = \text{erg}\cdot\text{cm}$, making $\sqrt{G}$ the explicit conversion factor between charge and mass. The expression is dimensionally valid without supplementary constants.

Repository Structure

```
cameron-unified-framework/
│
├── README.md                  ← this file
│
├── core/
│   ├── constants.py           ← single source of truth
│   ├── complex_charge.py      ← Q = q + im*sqrt(G)
│   └── redshift_formula.py    ← z = pi*G*rho*R^2/c^2
│
├── papers/
│   ├── cameron2012_dark_matter/
│   │   ├── notebook.ipynb     ← reproduces paper results
│   │   └── falsification.ipynb   ← what density breaks the fit?
│   ├── cameron2015_gravity/
│   │   ├── notebook.ipynb     ← (3,12) force, 32.4% fraction
│   │   └── precision_calc.py  ← 50dp arithmetic
│   └── cameron2018_redshift/
│       ├── notebook.ipynb     ← z=1100 reproduction
│       └── falsification.ipynb
│
├── predictions/
```

```
│   ├── 01_rotation_curves.ipynb   ← CGM profile vs SPARC data
│   ├── 02_mercury_perihelion.ipynb ← 43 arcsec/century
│   ├── 03_hubble_tension.ipynb    ← H0 from evanescence horizon
│   ├── 04_li7_abundance.ipynb     ← photodisintegration mechanism
│   └── 05_galaxy_morphology.ipynb ← size/shape from formation density
│
├── montecarlo/
│   ├── nuclear_vdw_baseline.py    ← uniform sphere, p-n sum=0
│   ├── nuclear_anchor_model.py    ← asymmetric quark distribution
│   └── nuclear_anchor_results.png ← 8-14σ attraction at 0.9-1.5 fm
│
└── disputes/
    ├── what_doesnt_work.md        ← honest list of open problems
    ├── llr_residual.ipynb         ← prediction: 10⁻¹⁸ mm, observed: 2mm
    ├── orbital_average.ipynb      ← full average: -3.62e-69 N vs Newton
    └── open_questions.md
```

What Is Open

These are not swept under the rug. They are the active research problems.

Gap 1 — The full orbital average. The (3,12) local mean gives -5.77×10^{-64} N (within factor 3 of Newton). The full orbit average gives -3.62×10^{-69} N (factor 50,000 too small). The physical mechanism that selects the (3,12) configuration — whether cosmological coherence, quantum ground state, or collective many-body effect — is not yet derived.

Gap 2 — The R^{-2} scaling. For gravity to be van der Waals, the force must scale as R^{-2} at macroscopic distances. Dipole-dipole forces scale as R^{-4}. The orbit averaging that converts one to the other is demonstrated numerically in Cameron (2015) but not analytically derived for 3D.

Gap 3 — The quark Monte Carlo. The relativistic γ factor for quarks moving at 0.99c has not been added. Expected to increase the signal ~49×, which would bring the potential well depth from ~1 MeV toward the empirical ~8 MeV/nucleon.

Gap 4 — G from first principles. The master equation $G \propto e^2 a_0^2/m_p^2$ gives G to within a factor of ~9 using the quark geometry. Closing this gap requires completing the Monte Carlo.

The Published Papers

All three are in *Physics Essays* and are the citable foundation:

1. **Cameron (2012)** — "Dark matter miscalculation from missing ISM shell mass" *Phys. Essays* **25**, 306

Phys. Essays **23**, 306
Claim: missing CGM mass, not exotic particles, explains flat rotation curves.

2. **Cameron (2015)** — "Is the force of gravity a manifestation of the electric force?"
   Phys. Essays **28**, 529
   Claim: (3,12) orbital configuration produces force within factor 3 of Newton.
3. **Cameron (2018)** — "Gravitational component of cosmological redshift"
   Phys. Essays **31**, [issue]
   Claim: $z = \pi G\rho R^2/c^2$ reproduces z=1100 at the Hubble radius.

Running the Calculations

Every number in the table above is reproducible:

```bash
git clone https://github.com/[your-repo]/cameron-unified-framework
cd cameron-unified-framework
pip install numpy scipy mpmath matplotlib

# Reproduce the (3,12) force (Cameron 2015 Table I):
python papers/cameron2015_gravity/precision_calc.py

# Run the quark Monte Carlo (Cameron 2015 anchor model):
python montecarlo/nuclear_anchor_model.py

# Reproduce z=1100 (Cameron 2018):
python core/redshift_formula.py
```

Results match the published values or the discrepancy is documented in `disputes/` with the reason explained.

Terminology

No invented terms are used in this repository. Every term below
is either standard physics or defined by an equation here.

Term used	Standard meaning	Defined by
Complex charge Q	$q + im\sqrt{G}$ in Gaussian units	Eq. above
Pseudo-Euclidean metric	$d^2=\alpha\cdot dx\_re^2+\beta\cdot dx\_im^2$	Cameron metric

Cross-tether	$\alpha/\beta = (m_e/m_p)^2$	Section I
PMV	Primary Momentum Vector	P_{re}, P_{im} pair
Kähler phase space	complex symplectic manifold	Section II.4
Evanescence horizon	radius where $z \to \infty$ in Cameron formula	Cameron 2018
(3,12) configuration	electron positions in 2D orbital model	Cameron 2015 Fig.1
van der Waals gravity	cross term of Q_1Q_2	this README

Terms not used here (from Gemini's version, no equation behind them):
exhaust, intake, umbilical, Phase Kinetics, nexus lattice,
flavor gradient, Heisenberg pressure, metric lubrication,
vortex drag, fluid tornado, unitary hum.

Acknowledgements

The three published papers are Donald Cameron's work.
Computational development, numerical verification, and mathematical formalization used Claude (Anthropic) as a computational tool.
Earlier framework development used Gemini (Google) for theoretical synthesis.

AI tools were used for arithmetic, pattern-matching across literature,
and code generation. Physical intuition, the original hypotheses,
and accountability for all claims belong to Donald Cameron.

"The number has to be able to come out wrong."
Every calculation in this repository was designed to fail.
The ones that didn't are the results.

For the non-specialist version of this framework, see:
The Story of Everything